Faculty of Engineering & Technology

Electrical & Computer Engineering Department

ENCS5341: MACHINE LEARNING AND DATA SCIENCE

Assignment 3: Predicting Weather and Time of Day from Images Using Machine Learning Models

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Abstract

This assignment investigates the use of machine learning models to predict weather conditions and time of day from images. Three models were evaluated: k-Nearest Neighbors (k-NN) as a baseline model, and Convolutional Neural Networks (CNN) and Random Forest (RF) as advanced models. The dataset was cleaned and analyzed using exploratory data analysis techniques, including class distribution analysis, descriptive statistics, and categorical correlation testing, to identify data imbalance and feature relationships. Model performance was evaluated using accuracy, joint accuracy, and F1-score. Hyperparameter tuning was conducted for CNN learning rates and RF tree counts to identify optimal configurations. Experimental results show that the CNN achieved the highest joint accuracy, indicating better simultaneous prediction of both tasks compared to k-NN and RF. Error analysis was performed by identifying misclassified samples and analyzing prediction patterns to understand model limitations. The results demonstrate the effectiveness of deep learning for multi-task image classification while highlighting challenges related to class imbalance, visual ambiguity, and dataset variability.

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Introduction

The Selected Task

The task selected in this assignment is the prediction of weather conditions and time of day from images. Each image is associated with two labels: a weather category and a time-of-day category. The problem is formulated as a multi-output image classification task, where a single input image is used to predict both labels simultaneously.

Explored Models

- **K-Nearest Neighbors (KNN):** Used as a baseline model due to its simplicity and minimal training requirements.
- **Convolutional Neural Network (CNN):** Applied to automatically extract spatial and visual features from images and perform multi-output classification.
- **Random Forest (RF):** Used as an ensemble learning method to compare classical machine learning performance with deep learning.

Evaluation Metrics

- **Accuracy:** Measures the percentage of correctly predicted labels for each task independently.
- **Joint Accuracy:** Measures the percentage of samples for which both weather and time labels are predicted correctly at the same time.
- **F1-score:** Provides a balanced measure between precision and recall, especially useful when class distributions are imbalanced.

Exploratory Data Analysis

Key EDA Steps

The dataset was loaded from the merged metadata file and inspected for data quality. Missing values were checked for all categorical attributes, including Weather, Time of Day, Season, Activity, Mood/Emotion, and Country. All columns contained zero missing values except Country, which had only one missing entry, indicating that the dataset is nearly complete and requires minimal cleaning.

Class distributions were analyzed for all categorical variables using bar plots to visualize category frequencies and percentages, enabling identification of class imbalance. Pairwise relationships between categorical variables were also examined using cross-tabulation heatmaps to explore label dependencies, with particular focus on the relationship between Weather and Time of Day.

Textual descriptions were evaluated by computing the length of each description and visualizing the distribution using a histogram. Descriptive statistics were calculated to summarize the minimum, maximum, mean, and variability of description lengths.

Descriptive Statistics and Visualizations

Missing Values

The missing value analysis showed:

- **Weather**: 0 missing values
- **Time of Day**: 0 missing values
- **Season**: 0 missing values
- **Activity**: 0 missing values
- **Mood/Emotion**: 0 missing values
- **Country**: 1 missing value

This indicates that the dataset is highly complete and suitable for supervised learning without major preprocessing.

Weather Distribution

The weather label distribution is highly imbalanced:

- **Sunny:** 592 samples
- **Cloudy:** 236 samples
- **Not Clear:** 103 samples
- **Snowy:** 90 samples
- **Clear:** 49 samples
- **Rainy:** 23 samples
- **Windy:** 3 samples

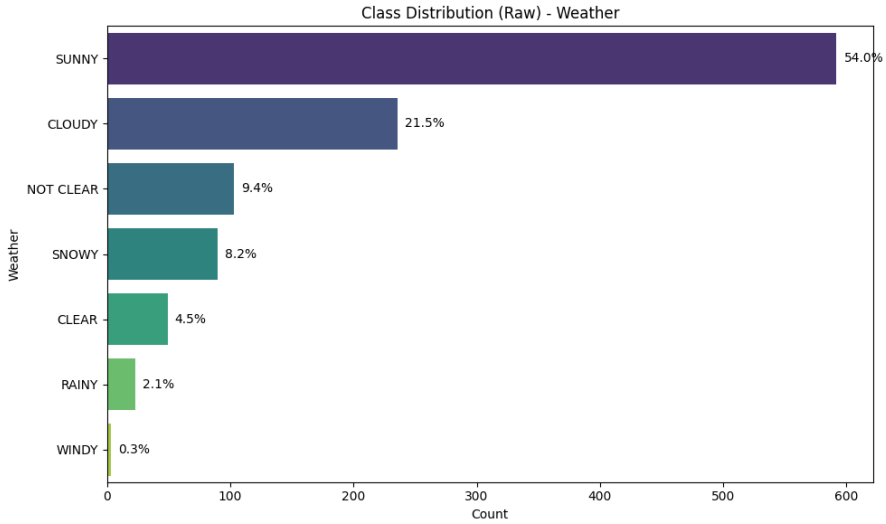

Figure 1: Weather Distribution plot.

Sunny images dominate the dataset, while Windy and Rainy have very few samples. This imbalance can negatively affect classification performance, especially for minority classes.

Time-of-Day Distribution

The time-of-day labels also show imbalance:

- **Afternoon:** 518 samples
- **Evening:** 311 samples
- **Morning:** 261 samples
- **Not Clear:** 6 samples

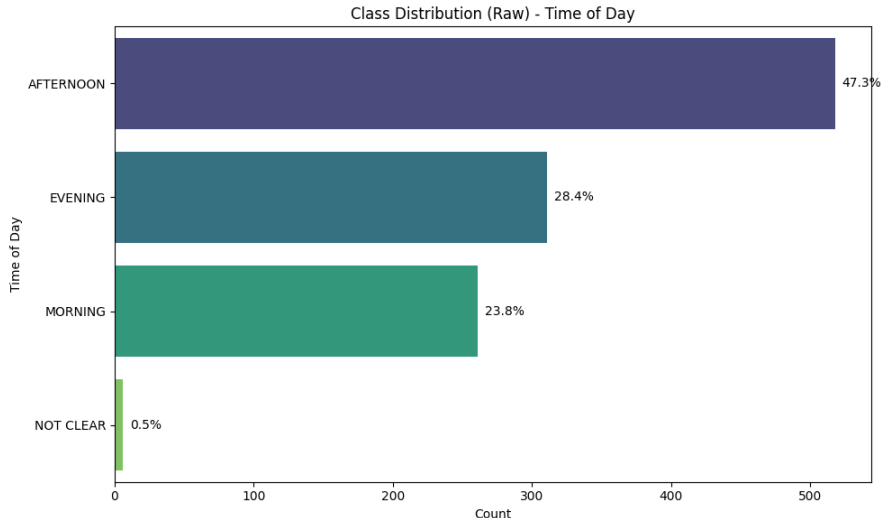

Figure 2: Time-of-Day Distribution plot

Most images were captured in the Afternoon and Evening, while the Not Clear category is extremely underrepresented which is better.

Weather vs Time-of-Day Relationship

The Weather vs Time-of-Day heatmap reveals several strong patterns:

- The largest cell occurs for Sunny & Afternoon (332 samples), indicating that most sunny images were taken in the afternoon.
- Sunny & Morning (169 samples) and Sunny & Evening (91 samples) also show high counts.
- Cloudy & Afternoon (121 samples) and Not Clear & Evening (80 samples) are additional strong combinations.

These gaps indicate sparse coverage for certain weather-time combinations, which makes it difficult for models to learn reliable patterns for these cases.

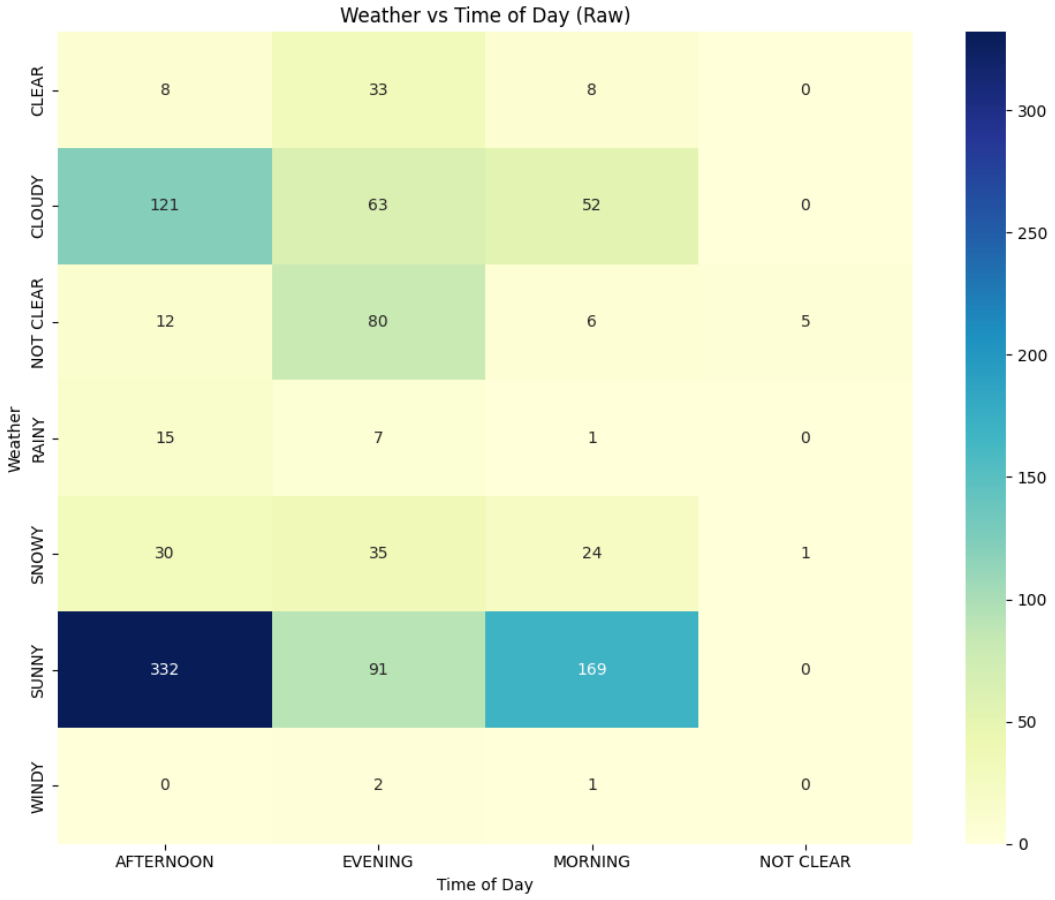

Figure 3: Weather vs Time-of-Day heatmap

Description Length Statistics

The textual description lengths show large variability:

- **Count:** 1,096 samples
- **Mean length:** 152 characters
- **Standard deviation:** 112 characters
- **Minimum:** 8 characters
- **Maximum:** 1,034 characters
- **Median (50%):** 135 characters

This indicates that descriptions range from very short to very detailed, which introduces variability in information quality across samples.

Observed Trends and Challenges

Several important trends and challenges were identified:

1. **Severe Class Imbalance**: Sunny and Afternoon dominate the dataset, while Windy, Rainy, and Not Clear categories contain very few samples. This increases the risk of biased predictions toward majority classes and reduced accuracy for minority classes.
2. **Sparse Weather-Time Combinations**: Many weather-time pairs have zero or near-zero samples. This limits the model's ability to learn joint patterns and contributes to lower joint prediction accuracy.

These findings motivated the selection of models capable of learning nonlinear patterns and handling high-dimensional image features, while also highlighting the need for careful evaluation using multiple metrics beyond accuracy.

Experiments and Results

Baseline Model k-Nearest Neighbors (k-NN)

The baseline model used was the k-Nearest Neighbors (k-NN) classifier. Since k-NN supports only single-output classification, the model was trained and evaluated separately for Weather and Time of Day prediction. Two values of k were tested: k = 1 and k = 3, using Euclidean distance.

Proposed Models

Two multi-output models were explored:

- **Convolutional Neural Network (CNN)**: A custom CNN was designed to learn spatial image features automatically. The network predicts Weather and Time of Day simultaneously using two output heads. Training used the Adam optimizer, dropout regularization, and early stopping.
- **Random Forest (RF):** A Random Forest classifier was trained using flattened image pixels. A MultiOutputClassifier wrapper was used to enable simultaneous prediction of Weather and Time of Day.

Hyperparameter Tuning

To improve performance, key hyperparameters were tuned:

CNN Hyperparameter

- Learning Rate tested: 0.0005, 0.001, 0.005, 0.01

Random Forest Hyperparameter

- Number of Trees tested: 50, 100, 300, 500

Each configuration was evaluated using Weather Accuracy, Time Accuracy, Joint Accuracy, and F1-score.

All Results Table

Model	Hyperparameter	Weather Acc	Time Acc	Joint Acc	Weather F1	Time F1
KNN	K = 1	0.4320	0.5030	-	0.2046	0.3603
KNN	K = 3	0.4379	0.5680	-	0.2103	0.3938
CNN	LR = 0.0005	0.6154	0.5799	0.3669	0.2297	0.4503
CNN	LR = 0.001	0.6391	0.5799	0.3669	0.2326	0.4705
CNN	LR = 0.005	0.6331	0.5621	0.3491	0.1576	0.4280
CNN	LR = 0.01	0.6272	0.5740	0.3669	0.1752	0.4470
RF	50 Trees	0.5740	0.5799	0.3432	0.2204	0.5113
RF	100 Trees	0.6036	0.5858	0.3491	0.2115	0.5168
RF	300 Trees	0.6272	0.5562	0.3491	0.2129	0.4783
RF	500 Trees	0.6154	0.5680	0.3373	0.2090	0.4874

Table 1: Model Performance Comparison

Performance Interpretation and Comparison

The CNN achieved the highest joint accuracy of 0.3669, making it the best overall model for simultaneous prediction of Weather and Time of Day. The optimal learning rate was 0.001, which also produced the highest Weather F1-score and Time F1-score among CNN configurations. The Random Forest model achieved competitive single-task accuracies but consistently lower joint accuracy compared to the CNN. Increasing the number of trees beyond 100 did not significantly improve joint performance and increased computational cost. The k-NN baseline performed substantially worse than both CNN and Random Forest, especially for Weather classification. This confirms that distance-based methods struggle with high-dimensional image data.

Overall, the CNN demonstrated superior ability to learn meaningful visual features and generalize across both classification tasks, justifying its selection as the best performing model.

Analysis

Error Analysis Methodology

Model errors were analyzed using a dedicated evaluation script that performs the following steps:

- The trained CNN generates predictions for the test dataset for both Weather and Time of Day.
- A prediction is marked as a joint error if either the weather label or the time-of-day label is incorrect.
- The total number of test samples and the total number of joint errors are computed.
- Error frequencies are calculated separately for Weather and Time-of-day labels.
- Showing some of the misclassified images.

Observations from Misclassified Samples

Visual inspection of misclassified images reveals several recurring patterns:

- Some images contain conflicting visual cues. For example, images with visible sunlight but also snow on the ground may be labeled as Sunny, while the model predicts Snowy due to the dominant white texture.
- Images captured during transitional lighting conditions (early morning or late afternoon) are sometimes confused with neighboring time categories.
- Low contrast, haze, or partial cloud coverage makes it difficult to distinguish between classes such as Clear, Cloudy, and Not Clear.

These examples indicate that the model often relies on dominant visual features that may not fully represent the semantic label.

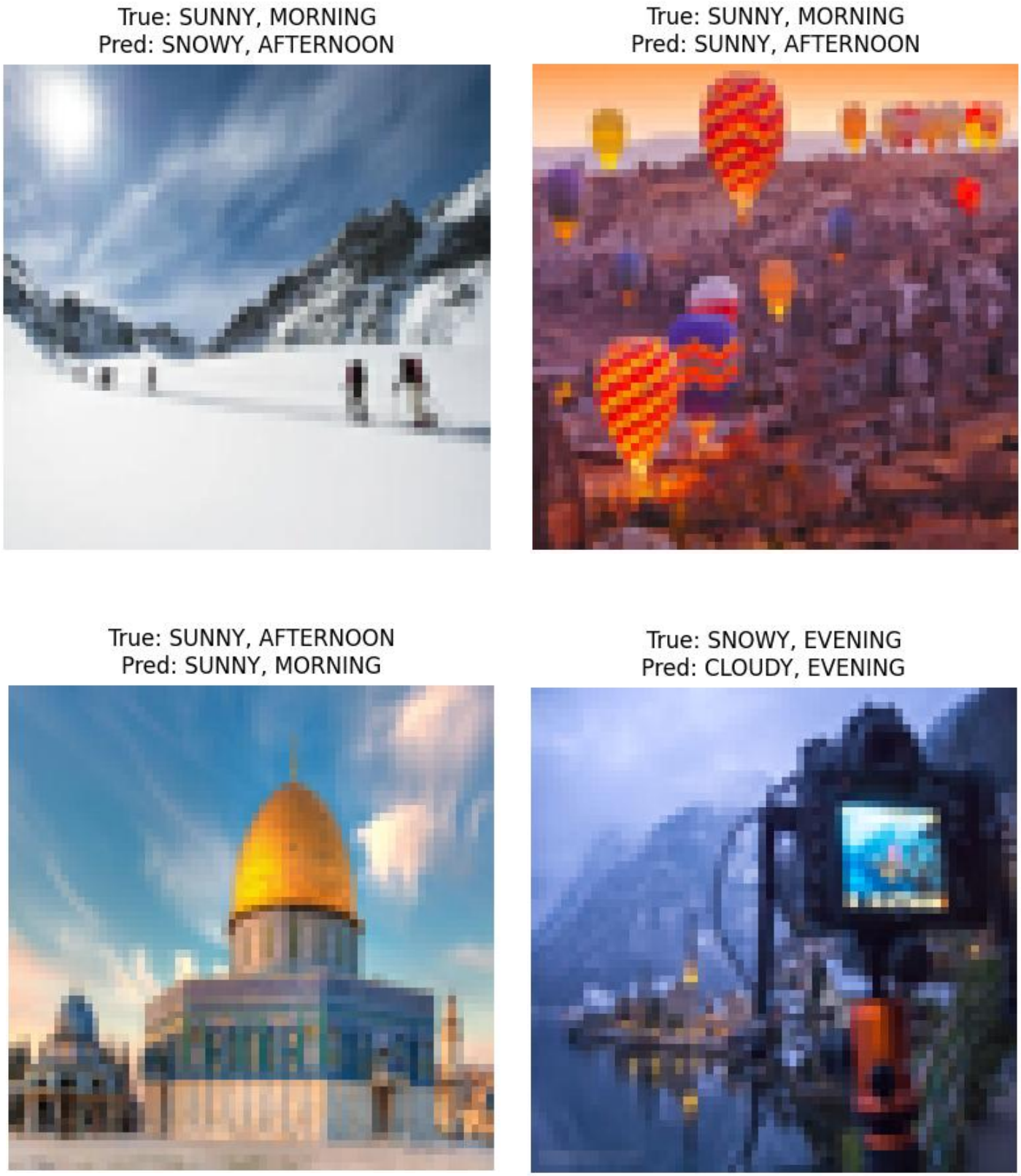

Figure 4: Examples of Misclassified Samples

Conclusion

This assignment addressed the task of jointly predicting Weather and Time of Day from images using three models: k-Nearest Neighbors (KNN), Convolutional Neural Network (CNN), and Random Forest (RF). KNN served as a baseline and achieved the lowest performance due to its reliance on flattened pixel vectors and the absence of spatial feature extraction. Random Forest achieved moderate individual accuracies but limited joint accuracy because spatial image structure is lost when pixels are flattened. The CNN achieved the highest joint accuracy (approximately 36-37%) and higher individual accuracies for both targets, demonstrating superior capability in learning spatial patterns from images. Hyperparameter tuning showed that moderate learning rates improved CNN stability, while increasing the number of trees in Random Forest provided only marginal gains.

Several limitations were observed. The dataset is imbalanced, with dominant classes such as Sunny and Afternoon and very few samples for rare categories such as Windy. Many images contain ambiguous visual cues, such as sunlight combined with snow or partial cloud coverage, leading to misclassification. The dataset size is limited for training deep models, increasing the risk of overfitting. In addition, joint accuracy is a strict metric, where a single mistake in either label counts as a full error. Furthermore, since the dataset was collected manually by multiple students, this introduces human labeling errors and occasional broken or inaccessible image links, reducing data reliability. Finally, KNN and RF suffer from poor feature representation due to loss of spatial information.

Future improvements include applying systematic data augmentation (rotation, flipping, brightness adjustment) to increase sample diversity and reduce overfitting, as well as balancing class distributions to mitigate bias toward dominant categories such as Sunny and Afternoon. Using higher-resolution images may preserve fine visual details that are important for distinguishing similar conditions. In addition, adopting pretrained CNN architectures through transfer learning can improve feature extraction when training data is limited.

Appendix: Code Execution and Data Cleaning Notes

The dataset was collected from multiple students and contained several formatting and encoding inconsistencies. Some Excel files could not be opened correctly due to UTF-8 encoding issues and inconsistent delimiters. These older files were inspected and excluded after verification.

Only a subset of the files required manual correction, while the remaining files were already usable and were kept unchanged. The following targeted fixes were applied where necessary:

- Quotation marks were added to image links to prevent commas in country names from being interpreted as extra columns.
- Empty columns created by repeated commas were removed.
- An extra non-required row was deleted to match the instructor's requested feature set.
- Extra trailing columns that caused inconsistencies during file merging were removed.

The file **merged_all_students.csv** provided and should be placed inside a folder named **Data_working_final**, as some of the original files required manual cleaning before merging.